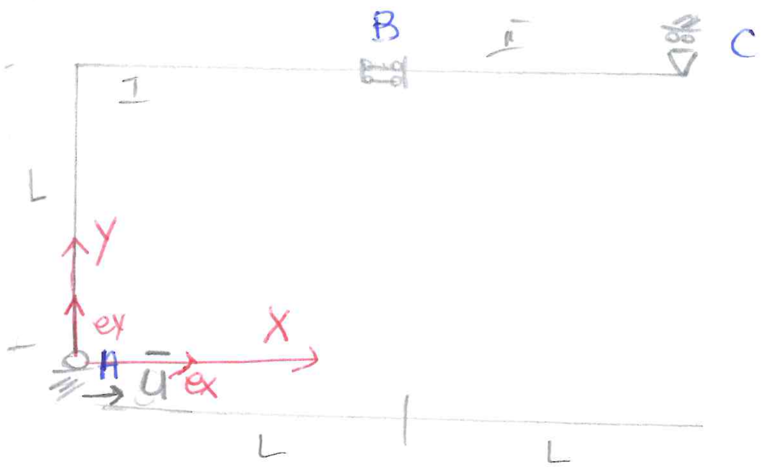
PROBLEMA CINEMATICO : 2 CORPI



1). Eq'ni cinematiche dei vincoli:

$$\underline{u}_A = \bar{u}$$

$$\Delta \underline{u}_B \cdot \underline{e}_B = 0 \rightarrow (\underline{u}_B'' - \underline{u}_B') \cdot \underline{e}_B \overset{\downarrow}{=} 0$$

$$\Delta \vartheta_B = 0 \rightarrow (\vartheta_B'' - \vartheta_B') = 0$$

$$\underline{u}_C \cdot \underline{e}_C = 0$$

2). scelta poli cinematici — corpo I : A

$$\underline{u}_P^{I} = \underline{u}_A + \vartheta^{I} \times (P - A)$$

corpo II : C

$$\underline{u}_P^{II} = \underline{u}_C + \vartheta^{II} \times (P - C)$$

3). Introduzione sistema di coordinate $\{A, x, y, z\}$

$A = (0, 0) \quad B = (L\underline{e}_x, L\underline{e}_y) \quad C = (2L\underline{e}_x, L\underline{e}_y) \quad P = (x\underline{e}_x, y\underline{e}_y)$

$$\underline{u}_P^{I} = u_{Ax}\,\underline{e}_x + u_{Ay}\,\underline{e}_y + \vartheta'\underline{e}_z \times (x\underline{e}_x + y\underline{e}_y)$$

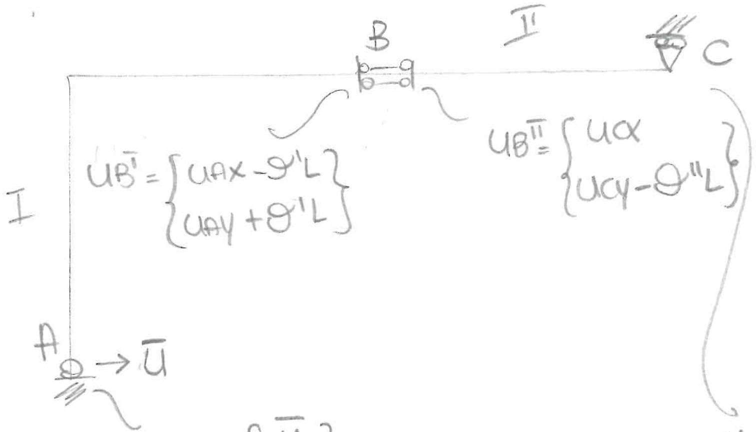
$$\overset{!}{=} u_{Ax}\,\underline{e}_x + u_{Ay}\,\underline{e}_y + (\vartheta' x\underline{e}_y - \vartheta' y\underline{e}_x)$$

$$\overset{!}{=} (u_{Ax} - \vartheta' y)\underline{e}_x + (u_{Ay} + \vartheta' x)\underline{e}_y$$

$$\underline{u}_P^{II} = u_{Cx}\,\underline{e}_x + u_{Cy}\,\underline{e}_y + \vartheta''\underline{e}_z \times [(x - 2L)\underline{e}_x + (y - L)\underline{e}_y]$$

$$\overset{!}{=} u_{Cx}\,\underline{e}_x + u_{Cy}\,\underline{e}_y + \vartheta'' x\underline{e}_y - 2L\vartheta''\underline{e}_y - \vartheta'' y\underline{e}_x + \vartheta'' L\underline{e}_x$$

$$\overset{!}{=} (u_{Cx} - \vartheta'' y + \vartheta'' L)\underline{e}_x + (u_{Cy} + \vartheta'' x - 2L\vartheta'')\underline{e}_y$$



$$\underline{u}_A = \begin{Bmatrix} \bar{u} \\ 0 \end{Bmatrix} \qquad \underline{u}_C = \begin{Bmatrix} u_{Cx} \\ u_{Cy} \end{Bmatrix}$$

$$\underline{u}_B^{I} = \begin{Bmatrix} u_{Ax} - \vartheta' L \\ u_{Ay} + \theta' L \end{Bmatrix} \qquad \underline{u}_B^{II} = \begin{Bmatrix} u_{Cx} \\ u_{Cy} - \theta'' L \end{Bmatrix}$$

$$\underline{u}_P^{I} = \begin{Bmatrix} u_{Ax} - \theta' y \\ u_{Ay} + \theta' x \end{Bmatrix}$$

$$\underline{u}_P^{II} = \begin{Bmatrix} u_{Cx} - \vartheta'' y + \theta'' L \\ u_{Cy} + \theta'' x - 2L\theta'' \end{Bmatrix}$$

$$\underline{u}_B^{I} = u_{Ax}\underline{e}_x + u_{Ay}\underline{e}_y + \theta' \times (B-A)$$

$$\overset{!}{=} (u_{Ax} - \theta^{I} L)\underline{e}_x + (u_{Ay} + \theta^{I} L)\underline{e}_y$$

$$\underline{u}_B^{II} = u_{Cx}\underline{e}_x + u_{Cy}\underline{e}_y + \theta'' \times (B-C)$$

$$\overset{!}{=} u_{Cx}\underline{e}_x + (u_{Cy} - \theta'' L)\underline{e}_y$$

$$\Delta \underline{u}_B = \underline{u}_B'' - \underline{u}_B' = (u_{Cx} - u_{Ax} + \theta' L)\underline{e}_x + (u_{Cy} - \theta'' L - u_{Ay} - \vartheta' L)\underline{e}_y$$

4). Eq'ni di vincolo in forma scalare

- $u_{Ax} = \bar{u}$
- $u_{Ay} = 0$
- $u_{Bx}'' - u_{Bx}' \Rightarrow u_{Cx} - u_{Ax} + \theta' L$
- $\Delta\vartheta_B = 0 \rightarrow \vartheta'' - \theta' = 0 \rightarrow \theta'' \overset{\checkmark}{=} \theta'$
- $\underline{u}_C \cdot \underline{e}_y \overset{\checkmark}{=} 0 \quad e_{Cy} = 0$

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & L & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & +1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}}_{A} \underbrace{\begin{Bmatrix} u_{Ax} \\ u_{Ay} \\ \vartheta' \\ u_{Cx} \\ u_{Cy} \\ \theta'' \end{Bmatrix}}_{q} = \underbrace{\begin{Bmatrix} \bar{u} \\ 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix}}_{S}$$

$R_3 \rightarrow R_3 + 1 \cdot R_1$
$R_4 \rightarrow R_4 + L R_3$

$\rightarrow rg(A) \overset{\checkmark}{=} 5$
$\rightarrow rg(A|S) \overset{\checkmark}{=} 5$

$\Big\}\ rg(A) = rg(A|S) \Rightarrow \exists$ sol'ne per R-C

• $rg(A) = rg(A|S) = 5$ $\rightarrow$ $n^{col} - rg(A) = 6 - 5 = 1$

$n = \# col = 6$

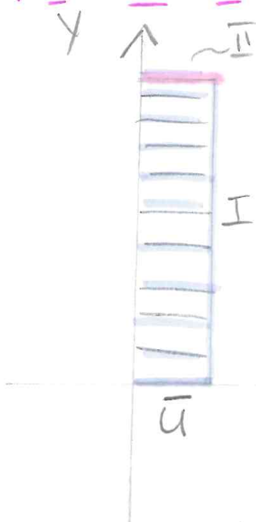
$\exists\, \infty^1$ soe^{ni} $\rightarrow$ La struttura è 1-volta labile

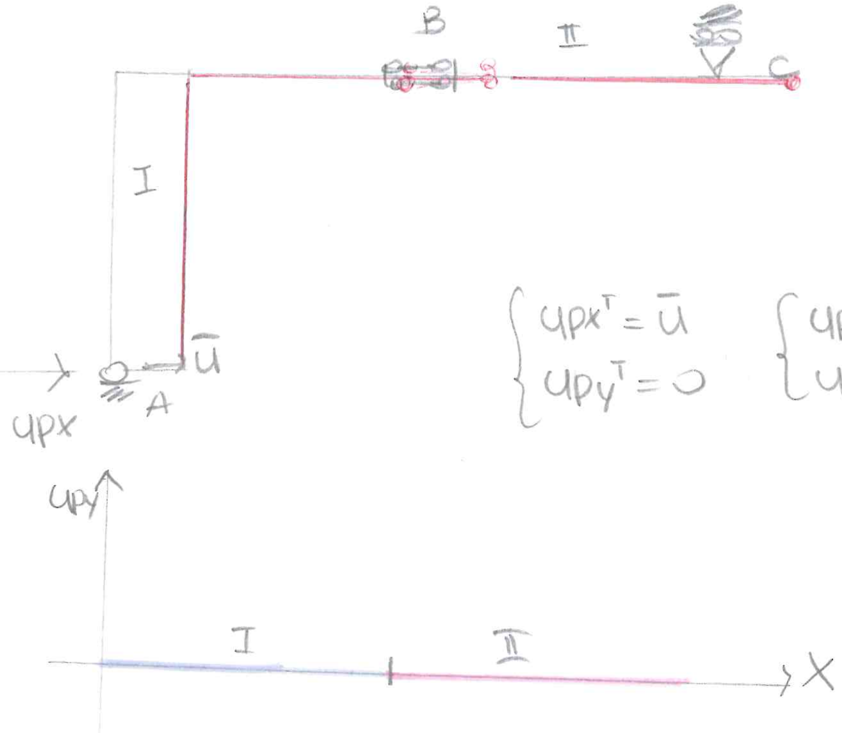
$$\begin{cases} u_{Ax} = \bar{u} \\ u_{Ay} = 0 \\ u_{Cx} - u_{Ax} + \theta^{I} L = 0 \\ \theta^{II} - \theta^{I} = 0 \\ u_{Cy} = 0 \end{cases} \rightarrow \begin{cases} u_{Ax} = \bar{u} \\ u_{Ay} = 0 \\ u_{Cx} = +\bar{u} - \theta^{I} L \\ \theta^{II} = \theta^{I} \\ u_{Cy} = 0 \end{cases} \rightarrow \theta^{I} \text{ parametro libero}$$

$\theta^{I} = t \in \mathbb{R}$ $\qquad$ $\underline{u} = \underline{u}_s + t\,\underline{u}_t$

5). Catene cinematiche (sovrapposizione degli effetti)

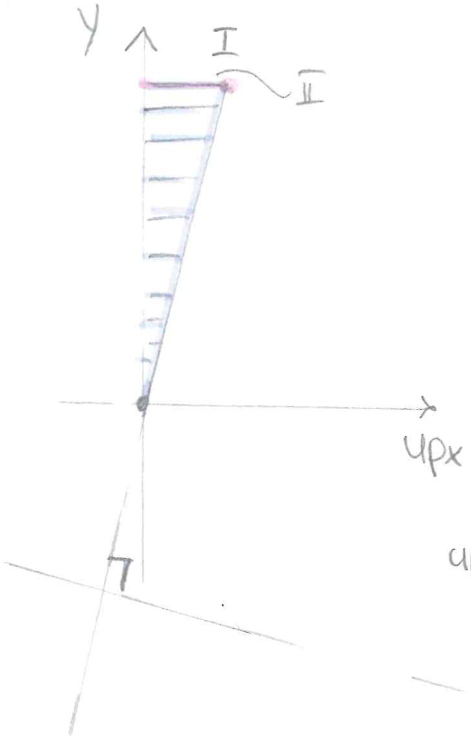
CASO 1 $\bar{u} \neq 0$ $t = 0$

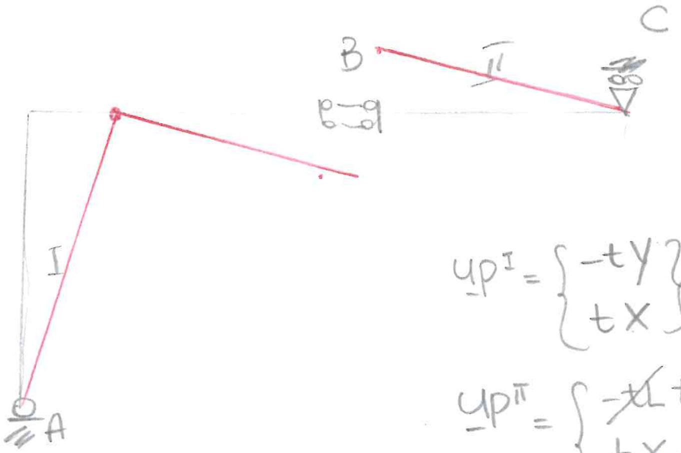




$$\begin{cases} u_{px}^{I} = \bar{u} \\ u_{py}^{I} = 0 \end{cases} \qquad \begin{cases} u_{px}^{II} = \bar{u} \\ u_{py}^{II} = 0 \end{cases}$$

CASO 2: $\bar{u}=0 \quad t\neq 0 \quad t>0$





$$\underline{u}p^{I} = \begin{Bmatrix} -ty \\ tx \end{Bmatrix}$$

$$\underline{u}p^{II} = \begin{Bmatrix} \cancel{-tL} + \cancel{tL} - ty \\ tX - 2Lt \end{Bmatrix}$$